

Vidvathama Ramesh

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Education

International Institute of Information Technology Hyderabad (IIIT-H)

Hyderabad, India

B.Tech in Electronics and Communication Engineering; CGPA: 9.01 / 10.0 (9.1+ in the last 3 sems)

July 2024 – Present

- Relevant Coursework: DSA, Statistical Methods in AI, Natural Language Processing, Processor Architecture, Digital Systems and Logic, VLSI Design, Information Theory, Probability and Random Processes, Systems Thinking, Functional Analysis.

Research Experience

Research Intern — TCS Research

May 2026 – July 2026

Mentor: Jeripothula Prudviraj

Hyderabad, India

- Architected deep neural networks that extract high-level aesthetic and stylistic features as state representations for autonomous RL agents, optimizing policy-gradient reward functions for iterative generative decision-making.
- Engineered a human-AI co-creation framework integrating CV and RL for real-time agents that adapt to human creative intent; preparing submissions to **NeurIPS 2026** (Creative AI Track: Agency) and **ACM CHI 2026**.

Undergraduate Researcher — Robotics Research Center (RRC)

May 2025 – Present

Supervisor: Prof. Sourav Garg

Hyderabad, India

- Integrating Vision-Language Actions/Models (VLAs/VLMs) with topological graph structuring for robust navigation in unstructured environments, and developing 3D-vision + RL models for LEAP Hand dexterous manipulation.

Research Intern — Indian Institute of Technology Bombay (IIT B)

May 2025 – Aug 2025

Supervisor: Prof. Vikram Bansal

Mumbai, India

- Implemented a TransCrowd Transformer pipeline for high-density crowd counting, integrating spatio-temporal features to improve density estimation accuracy across large-scale datasets.

Technical Projects

Adaptive Semantic Pipeline for LLM Routing | *DistilBERT, PyTorch, LightGBM, spaCy*

Jan. 2026

- Architected an 8-stage LLM routing pipeline decomposing prompts into a semantic DAG (spaCy, SentenceTransformer, constituency parsing) and routing sub-tasks to strong/weak models via a 27-feature LightGBM router with BFS strong-inheritance (85.66% accuracy, 88.04% F1), cutting inference cost $\sim 49\%$ at 131 ms mean latency.
- Fine-tuned a dual-head DistilBERT (11-class intent + complexity regression; 83.32% accuracy, 85.32% F1) and built an async DAG engine with speculative execution, semantic caching, and a context blackboard (60–80% token savings), plus a 3-layer safety, NLI, and LLM-as-a-Judge stack.

CanvasMind — Multi-Agent Inference-Time RL for Creative AI | *FastAPI, Azure OpenAI, Python*

May 2026

- Built a multi-agent creative-AI system where generative agents co-paint a shared canvas under an inference-time RL layer using best-of-N search, a batched reward model, a persistent UCB1 bandit, and reward-triggered Reflexion self-correction.
- Modeled agent interactions with misaligned rewards in a general-sum game and applied Shapley, empowerment, and Goodhart diagnostics to identify and correct RL defects.

PolyQuery — Multi-Model Relational Algebra Engine | *Systems, Concurrency*

Nov. 2025

- Built a relational algebra engine integrating graph processing and high-dimensional vector search over a custom block-based buffer manager with external-memory algorithms (external merge sort, hash, B-Tree).
- Implemented multi-threaded transactions with Wait-Die concurrency control and page-granular shared/exclusive locking (deadlock-free), disk-backed BFS path queries, and KNN search (Euclidean/Cosine) via a bounded min-heap under a fixed 10-page memory ceiling.

POSIX-Core C Shell | *C, Unix Syscalls, Signals, Valgrind*

Aug. 2025

- Developed a modular `execvp`-based Unix shell with foreground/background execution, unlimited pipelines, full I/O redirection, and `.myshrc` alias resolution; verified leak-free execution with Valgrind.
- Implemented 12+ built-ins (hop, reveal, seek, log, proclore, activities, fg/bg, neonate, and an iMan TCP man-page fetcher) with robust SIGINT/SIGTSTP/SIGCHLD handling and asynchronous child reaping.

Technical Skills

Languages: C, C++, Python, SQL, JavaScript, Bash, Assembly x86, MATLAB, LaTeX

Machine Learning: PyTorch, TensorFlow, Scikit-Learn, HuggingFace, LightGBM, OpenCV, NumPy, Pandas, spaCy

Frameworks & Systems: FastAPI, Flask, Next.js, React, Node.js, LangGraph, ROS 2, Isaac Gym, Linux/Unix, POSIX

Tools & Databases: Git, Docker, PostgreSQL, MySQL, Valgrind, Azure OpenAI

Achievements & Honors

Hackathon Honors: 1st Prize, SUD Life Bank Hackathon; Top 10 Finalist, Lloyds IP&I Hackathon (2025) — voice-first IVR replacement using RAG (Whisper, GPT-4o, FAISS) to ground responses in policy documents and auto-file tickets.

Dean's List: Awarded for Academic Excellence — Top 5% of batch (Spring UG 2) and Top 10% (Monsoon UG 3).